

$$\Rightarrow T_n = n(n+1)^2 = n^3 + 2n^2 + n$$

Let S_n denote the sum to n terms of the given series. Then,

$$S_n = \sum_{k=1}^n T_k = \sum_{k=1}^n (k^3 + 2k^2 + k)$$

$$\Rightarrow S_n = \sum_{k=1}^n k^3 + 2 \sum_{k=1}^n k^2 + \sum_{k=1}^n k$$

$$\Rightarrow S_n = \left\{ \frac{n(n+1)}{2} \right\}^2 + 2 \left\{ \frac{n(n+1)(2n+1)}{6} \right\} + \left\{ \frac{n(n+1)}{2} \right\}$$

$$\Rightarrow S_n = \frac{n(n+1)}{2} \left\{ \frac{n(n+1)}{2} + \frac{2}{3}(2n+1) + 1 \right\} = \frac{n(n+1)}{2} \left\{ \frac{3n^2 + 3n + 8n + 4 + 6}{6} \right\}$$

$$\Rightarrow S_n = \frac{n(n+1)}{2} \left\{ \frac{3n^2 + 11n + 10}{6} \right\} = \frac{n(n+1)(n+2)(3n+5)}{12}$$

EXAMPLE 5 Sum the series $3.8 + 6.11 + 9.14 + \dots$ to n terms.

[NCERT]

SOLUTION Let T_n be the n th term of the given series. Then,

$$T_n = (\text{nth term of } 3, 6, 9, \dots) \times (\text{nth term of } 8, 11, 14, \dots)$$

$$\Rightarrow T_n = [3 + (n-1) \times 3] \times [8 + (n-1) \times 3] = 3n(3n+5) = 9n^2 + 15n$$

Let S_n denote the sum to n terms of the given series. Then,

$$S_n = \sum_{k=1}^n T_k = \sum_{k=1}^n (9k^2 + 15k) = 9 \sum_{k=1}^n k^2 + 15 \sum_{k=1}^n k$$

$$\Rightarrow S_n = 9 \left\{ \frac{n(n+1)(2n+1)}{6} \right\} + 15 \left\{ \frac{n(n+1)}{2} \right\} = \frac{3}{2} n(n+1)[2n+1+5] = 3n(n+1)(n+3)$$

EXAMPLE 6 Find the sum of n terms of the series whose n th term is

(i) $2n^2 - 3n + 5$

(ii) $n^2 + 2^n$

[NCERT]

SOLUTION (i) We have, $T_n = 2n^2 - 3n + 5$.

Let S_n denote the sum of n terms of the series whose n th term is T_n . Then,

$$S_n = \sum_{k=1}^n T_k = \sum_{k=1}^n (2k^2 - 3k + 5) = 2 \sum_{k=1}^n k^2 - 3 \sum_{k=1}^n k + \sum_{k=1}^n 5$$

$$\Rightarrow S_n = 2 \left\{ \frac{n(n+1)(2n+1)}{6} \right\} - 3 \left\{ \frac{n(n+1)}{2} \right\} + 5n$$

$$\Rightarrow S_n = \frac{n}{6} \left\{ 2(n+1)(2n+1) - 9(n+1) + 30 \right\} = \frac{n}{6} (4n^2 + 6n + 2 - 9n - 9 + 30)$$

$$\Rightarrow S_n = \frac{n}{6} (4n^2 - 3n + 23)$$

(ii) We have, $T_n = n^2 + 2^n$

Let S_n denote the sum of n terms of the series having T_n as its n th term. Then,

$$S_n = \sum_{k=1}^n T_k = \sum_{k=1}^n (k^2 + 2^k) = \sum_{k=1}^n k^2 + \sum_{k=1}^n 2^k$$

$$\Rightarrow S_n = \frac{n(n+1)(2n+1)}{6} + (2^1 + 2^2 + 2^3 + \dots + 2^n)$$

$$\Rightarrow S_n = \frac{n(n+1)(2n+1)}{6} + 2 \left(\frac{2^n - 1}{2 - 1} \right) = \frac{n(n+1)(2n+1)}{6} + 2(2^n - 1)$$

EXAMPLE 7 Find the sum of the following series to n terms:

$$\frac{1^3}{1} + \frac{1^3 + 2^3}{1 + 3} + \frac{1^3 + 2^3 + 3^3}{1 + 3 + 5} + \dots$$

[NCERT]

SOLUTION Let T_n be the n th term of the given series. Then,

$$T_n = \frac{1^3 + 2^3 + 3^3 + \dots + n^3}{1 + 3 + 5 + \dots + (2n-1)} = \frac{\left\{ \frac{n(n+1)}{2} \right\}^2}{\frac{n}{2} \{1 + (2n-1)\}} = \frac{(n+1)^2}{4} = \frac{1}{4}(n^2 + 2n + 1)$$

Let S_n denote the sum of n terms of the given series. Then,

$$S_n = \sum_{k=1}^n T_k = \sum_{k=1}^n \frac{1}{4}(k^2 + 2k + 1) = \frac{1}{4} \left\{ \sum_{k=1}^n k^2 + 2 \sum_{k=1}^n k + \sum_{k=1}^n 1 \right\}$$

$$\Rightarrow S_n = \frac{1}{4} \left\{ \frac{n(n+1)(2n+1)}{6} + 2 \left(\frac{n(n+1)}{2} \right) + n \right\} = \frac{n}{24}(2n^2 + 9n + 13)$$

EXAMPLE 8 Show that: $\frac{1 \times 2^2 + 2 \times 3^2 + \dots + n \times (n+1)^2}{1^2 \times 2 + 2^2 \times 3 + \dots + n^2 \times (n+1)} = \frac{3n+5}{3n+1}$.

[NCERT]

SOLUTION Let T_n and T'_n be the n th terms of the series in numerator and denominator of LHS. Then,

$$T_n = n \text{th term of the series } 1 \times 2^2 + 2 \times 3^2 + \dots + n \times (n+1)^2$$

$$T_n = n(n+1)^2 = n^3 + 2n^2 + n$$

and, $T'_n = n \text{th term of the series } 1^2 \times 2 + 2^2 \times 3 + \dots + n^2 \times (n+1)$

$$T'_n = n^2(n+1) = n^3 + n^2$$

$$\therefore \text{LHS} = \frac{\sum_{k=1}^n T_k}{\sum_{k=1}^n T'_k} = \frac{\sum_{k=1}^n (k^3 + 2k^2 + k)}{\sum_{k=1}^n (k^3 + k^2)} = \frac{\sum_{k=1}^n k^3 + 2 \sum_{k=1}^n k^2 + \sum_{k=1}^n k}{\sum_{k=1}^n k^3 + \sum_{k=1}^n k^2}$$

$$\Rightarrow \text{LHS} = \frac{\left\{ \frac{n(n+1)}{2} \right\}^2 + 2 \left\{ \frac{n(n+1)(2n+1)}{6} \right\} + \left\{ \frac{n(n+1)}{2} \right\}}{\left\{ \frac{n(n+1)}{2} \right\}^2 + \left\{ \frac{n(n+1)(2n+1)}{6} \right\}}$$

$$\Rightarrow \text{LHS} = \frac{\frac{n(n+1)}{2} \left\{ \frac{n(n+1)}{2} + \frac{2(2n+1)}{3} + 1 \right\}}{\frac{n(n+1)}{2} \left\{ \frac{n(n+1)}{2} + \frac{(2n+1)}{3} \right\}}$$

$$\Rightarrow \text{LHS} = \frac{\frac{3n^2 + 3n + 8n + 4 + 6}{6}}{\frac{3n^2 + 3n + 4n + 2}{6}} = \frac{3n^2 + 11n + 10}{3n^2 + 7n + 2} = \frac{(3n+5)(n+2)}{(3n+1)(n+2)} = \frac{3n+5}{3n+1} = \text{RHS}$$

EXAMPLE 9 If S_1, S_2, S_3 are the sums of first n natural numbers, their squares, their cubes respectively, show that $9S_2^2 = S_3(1 + 8S_1)$. [NCERT]

SOLUTION We have,

$$S_1 = \sum_{k=1}^n k = 1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

$$S_2 = \sum_{k=1}^n k^2 = 1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

and, $S_3 = \sum_{k=1}^n k^3 = 1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$

$$\therefore 9S_2^2 = 9 \left\{ \frac{n(n+1)(2n+1)}{6} \right\}^2 = \frac{9}{36} \left\{ n(n+1)(2n+1) \right\}^2 = \frac{1}{4} \left\{ n(n+1)(2n+1) \right\}^2 \dots (i)$$

and, $S_3(1 + 8S_1) = \left\{ \frac{n(n+1)}{2} \right\}^2 \left\{ 1 + 8 \times \frac{n(n+1)}{2} \right\} = \left\{ \frac{n(n+1)}{2} \right\}^2 (4n^2 + 4n + 1)$

$$\Rightarrow S_3(1 + 8S_1) = \frac{n^2(n+1)^2(2n+1)^2}{4} = \frac{1}{4} \left\{ n(n+1)(2n+1) \right\}^2 \dots (ii)$$

From (i) and (ii), we obtain

$$9S_2^2 = S_3(1 + 8S_1).$$

EXAMPLE 10 Find the sum to n terms of the series: $1^2 + (1^2 + 2^2) + (1^2 + 2^2 + 3^2) + \dots$ [NCERT]

SOLUTION Let T_n be the n th term of the given series. Then,

$$T_n = 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} = \frac{1}{6}(2n^3 + 3n^2 + n)$$

Let S_n be the sum to n terms of the given series. Then,

$$S_n = \sum_{k=1}^n T_k = \sum_{k=1}^n \frac{1}{6}(2k^3 + 3k^2 + k)$$

$$\Rightarrow S_n = \frac{2}{6} \left(\sum_{k=1}^n k^3 \right) + \frac{3}{6} \sum_{k=1}^n k^2 + \frac{1}{6} \sum_{k=1}^n k$$

$$\Rightarrow S_n = \frac{1}{3} \left(\sum_{k=1}^n k^3 \right) + \frac{1}{2} \left(\sum_{k=1}^n k^2 \right) + \frac{1}{6} \left(\sum_{k=1}^n k \right)$$

$$\Rightarrow S_n = \frac{1}{3} \left\{ \frac{n(n+1)}{2} \right\}^2 + \frac{1}{2} \left\{ \frac{n(n+1)(2n+1)}{6} \right\} + \frac{1}{6} \left\{ \frac{n(n+1)}{2} \right\}$$

$$\Rightarrow S_n = \frac{n(n+1)}{12} \left\{ n(n+1) + (2n+1) + 1 \right\} = \frac{n(n+1)}{12} (n^2 + 3n + 2) = \frac{n}{12} (n+1)^2 (n+2).$$

LEVEL-2

EXAMPLE 11 The sequence N of natural numbers is divided into classes as follows:

$$\begin{array}{cccccc} & & 1 & 2 & & \\ & 3 & 4 & 5 & 6 & \\ 7 & 8 & 9 & 10 & 11 & 12 \end{array}$$

.....
.....

Show that the sum of the numbers in n th row is $n(2n^2 + 1)$.

SOLUTION Since the first row consists of 2 natural numbers, second row 4 natural numbers, third row 6 natural numbers and so on. So, the total number of natural numbers in n th row is $2n$. Now,

Total number of natural numbers upto the end of n th row

$$= 2 + 4 + 6 + \dots + 2n = 2(1 + 2 + \dots + n) = \frac{2n(n+1)}{2} = n(n+1).$$

$\therefore$ Total number of natural numbers upto the end of $(n-1)$ th row $= (n-1)(n-1+1) = n(n-1)$.

Let S_n denote the sum of first n natural numbers. Then,

Sum of the natural numbers in n th row

= Sum of the natural numbers upto the end of n th row

– Sum of the natural numbers upto the end of $(n-1)$ th row

$$= S_{n(n+1)} - S_{n(n-1)}$$

$$= S_m - S_p, \text{ where } m = n(n+1), p = n(n-1).$$

$$= \frac{m(m+1)}{2} - \frac{p(p+1)}{2}$$

$$\left[\text{Using: } S_n = \frac{n(n+1)}{2} \right]$$

$$= \frac{n(n+1)\{n(n+1)+1\}}{2} - \frac{n(n-1)\{n(n-1)+1\}}{2}$$

$$= \frac{n}{2} \left\{ (n+1)(n^2+n+1) - (n-1)(n^2-n+1) \right\} = \frac{n}{2} (4n^2+2) = n(2n^2+1).$$

EXAMPLE 12 If $S_k = \frac{1+2+\dots+k}{k}$, find the value of $S_1^2 + S_2^2 + \dots + S_n^2$.

SOLUTION We have,

$$S_k = \frac{1+2+\dots+k}{k} = \frac{k(k+1)}{2k} = \frac{k+1}{2}$$

$$\begin{aligned} \therefore S_1^2 + S_2^2 + \dots + S_n^2 &= \sum_{k=1}^n S_k^2 = \sum_{k=1}^n \left(\frac{k+1}{2} \right)^2 = \frac{1}{4} \sum_{k=1}^n (k+1)^2 \\ &= \frac{1}{4} \left\{ 2^2 + 3^2 + \dots + (n+1)^2 \right\} = \frac{1}{4} \left\{ 1^2 + 2^2 + \dots + (n+1)^2 - 1^2 \right\} \\ &= \frac{1}{4} \left\{ \frac{(n+1)(n+1+1)\{2(n+1)+1\}}{6} - 1 \right\} \\ &= \frac{1}{4} \left\{ \frac{(n+1)(n+2)(2n+3)}{6} - 1 \right\} = \frac{n}{24} (2n^2 + 9n + 13) \end{aligned}$$

EXAMPLE 13 Sum to n terms the series: $1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \dots$

SOLUTION Clearly, n th term of the given series is negative or positive according as n is even or odd respectively.

CASE I When n is even : In this case the given series is

$$\begin{aligned} &1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \dots + (n-1)^2 - n^2 \\ &= (1^2 - 2^2) + (3^2 - 4^2) + (5^2 - 6^2) + \dots + ((n-1)^2 - n^2) \end{aligned}$$

$$\begin{aligned}
 &= (1-2)(1+2) + (3-4)(3+4) + (5-6)(5+6) + \dots + \{(n-1)-(n)\}(n-1+n) \\
 &= -(1+2+3+4+\dots+(n-1)+n) = -\frac{n(n+1)}{2}.
 \end{aligned}$$

CASE II When n is odd : In this case the given series is

$$\begin{aligned}
 &(1^2 - 2^2) + (3^2 - 4^2) + \dots + \{(n-2)^2 - (n-1)^2\} + n^2 \\
 &= (1-2)(1+2) + (3-4)(3+4) + \dots + \{(n-2)-(n-1)\}\{(n-2)+(n-1)\} + n^2 \\
 &= -[1+2+3+4+\dots+(n-2)+(n-1)] + n^2 = -\frac{(n-1)(n-1+1)}{2} + n^2 = \frac{n(n+1)}{2}.
 \end{aligned}$$

EXAMPLE 14 Find the sum of all possible products of the first n natural numbers taken two by two.

SOLUTION We have,

$$(x_1 + x_2 + \dots + x_n)^2 = (x_1^2 + x_2^2 + \dots + x_n^2) + 2 \text{ (Sum of all possible products taken two at a time)}$$

$$\text{or, } \left(\sum_{i=1}^n x_i \right)^2 = \left(\sum_{i=1}^n x_i^2 \right) + 2 \left(\sum_{i=1, i < j}^n \sum_{j=1}^n x_i x_j \right)$$

$$\Rightarrow \sum_{i=1, i < j}^n \sum_{j=1}^n x_i x_j = \frac{1}{2} \left\{ \left(\sum_{i=1}^n x_i \right)^2 - \left(\sum_{j=1}^n x_j^2 \right) \right\}$$

$$\begin{aligned}
 \therefore \text{ Required sum} &= \frac{1}{2} \left\{ \left(\sum_{k=1}^n k \right)^2 - \left(\sum_{k=1}^n k^2 \right) \right\} = \frac{1}{2} \left[\left\{ \frac{n(n+1)}{2} \right\}^2 - \frac{n(n+1)(2n+1)}{6} \right] \\
 &= \frac{1}{2} \left[\frac{n(n+1)}{2} \left\{ \frac{n(n+1)}{2} - \frac{2n+1}{3} \right\} \right] = \frac{n(n+1)}{4} \left\{ \frac{3n^2 + 3n - 4n - 2}{6} \right\} \\
 &= \frac{n(n+1)(3n^2 - n - 2)}{24} = \frac{n(n+1)(n-1)(3n+2)}{24}
 \end{aligned}$$

EXAMPLE 15 Find the sum of the series: $1.n + 2.(n-1) + 3.(n-2) + \dots + (n-1).2 + n.1$.

SOLUTION Let T_r be the r th term of the given series. Then,

$$T_r = r\{n-(r-1)\} = r(n-r+1) = r\{(n+1)-r\} = (n+1)r - r^2$$

$$\therefore \text{ Required sum} = \sum_{r=1}^n T_r$$

$$= \sum_{r=1}^n \{(n+1)r - r^2\}$$

$$= (n+1) \left(\sum_{r=1}^n r \right) - \left(\sum_{r=1}^n r^2 \right)$$

$$= (n+1) \frac{n(n+1)}{2} - \frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)(n+2)}{6}$$

EXAMPLE 16 Find the sum of the series

$$(3^3 - 2^3) + (5^3 - 4^3) + (7^3 - 6^3) + \dots \text{ to } n \text{ terms}$$

[NCERT EXEMPLAR]**SOLUTION** Let S be the sum of the given series. Then,

$$S = (3^3 - 2^3) + (5^3 - 4^3) + (7^3 - 6^3) + \dots \text{ to } n \text{ terms.}$$

$$\Rightarrow S = \sum_{r=1}^n \left\{ (2r+1)^3 - (2r)^3 \right\}$$

$$\Rightarrow S = \sum_{r=1}^n \left\{ (2r+1) - (2r) \right\} \left\{ (2r+1)^2 + (2r+1)(2r) + (2r)^2 \right\}$$

$$\Rightarrow S = \sum_{r=1}^n (12r^2 + 6r + 1)$$

$$\Rightarrow S = 12 \sum_{r=1}^n r^2 + 6 \sum_{r=1}^n r + \sum_{r=1}^n 1$$

$$\Rightarrow S = 12 \times \frac{n(n+1)(2n+1)}{6} + 6 \times \frac{n(n+1)}{2} + n$$

$$\Rightarrow S = 2n(n+1)(2n+1) + 3n(n+1) + n$$

$$\Rightarrow S = n(4n^2 + 6n + 2 + 3n + 3 + 1) = n(4n^2 + 9n + 6)$$

EXERCISE 21.1**LEVEL-1**Find the sum of the following series to n terms: (1-7)

1. $1^3 + 3^3 + 5^3 + 7^3 + \dots$

2. $2^3 + 4^3 + 6^3 + 8^3 + \dots$

3. $1.25 + 2.36 + 3.47 + \dots$

4. $1.24 + 2.37 + 3.410 + \dots$

5. $1 + (1+2) + (1+2+3) + (1+2+3+4) + \dots$

6. $1 \times 2 + 2 \times 3 + 3 \times 4 + 4 \times 5 + \dots$

[NCERT]

7. $3 \times 1^2 + 5 \times 2^2 + 7 \times 3^2 + \dots$

[NCERT]8. Find the sum of the series whose n th term is:

(i) $2n^3 + 3n^2 - 1$

(ii) $n^3 - 3^n$

(iii) $n(n+1)(n+4)$ **[NCERT]**

(iv) $(2n-1)^2$

[NCERT]9. Find the 20th term and the sum of 20 terms of the series:

$$2 \times 4 + 4 \times 6 + 6 \times 8 + \dots$$

[NCERT]**ANSWERS**

1. $n^2(2n^2 - 1)$

2. $2[n(n+1)]^2$

3. $\frac{n}{12}(n+1)(3n^2 + 23n + 34)$

4. $\frac{n}{12}(n+1)(9n^2 + 25n + 14)$

5. $\frac{n(n+1)(n+2)}{6}$

6. $\frac{n}{3}(n+1)(n+2)$

7. $\frac{n}{6}(n+1)(3n^2 + 5n + 1)$

8. (i) $\frac{n}{2}(n^3 + 4n^2 + 4n - 1)$

(ii) $\left\{ \frac{n(n+1)}{2} \right\}^2 - \frac{3}{2}(3^n - 1)$

(iii) $\frac{n(n+1)}{12}(3n^2 + 23n + 34)$

(iv) $\frac{n}{3}(2n+1)(2n-1)$

9. 1680, 12320

HINTS TO NCERT & SELECTED PROBLEMS

1. Clearly, $T_n = (2n)^2 = 4n^2$

$$\therefore S_n = \sum_{k=1}^n T_k = 4 \sum_{k=1}^n k^2 = 4 \frac{n(n+1)(2n+1)}{6} = \frac{2}{3} n(n+1)(2n+1)$$

2. Clearly, $T_n = (2n-1)^3$

$$\therefore S_n = \sum_{k=1}^n T_k = \sum_{k=1}^n (2k-1)^3 = 8 \sum_{k=1}^n k^3 - 12 \sum_{k=1}^n k^2 + 6 \sum_{k=1}^n k - \sum_{k=1}^n 1$$

3. Clearly, $T_n = (2n)^3$

$$\therefore S_n = \sum_{k=1}^n T_k = \sum_{k=1}^n (2k)^3 = 8 \sum_{k=1}^n k^3 = 8 \left\{ \frac{n(n+1)}{2} \right\}^2 = 2n^2 (n+1)^2$$

8. Let T_r be the r^{th} term of the given series and S_n denote the sum of its n terms. Then,

$$T_r = r(r+1), \quad r=1, 2, 3, \dots$$

$$\therefore S_n = \sum_{r=1}^n T_r = \sum_{r=1}^n r(r+1)$$

$$\Rightarrow S_n = \sum_{r=1}^n (r^2 + r) = \sum_{r=1}^n r^2 + \sum_{r=1}^n r$$

$$\Rightarrow S_n = \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} = \frac{n(n+1)(2n+4)}{6} = \frac{n(n+1)(n+2)}{3}$$

9. Let T_r be the r^{th} term of the series and S_n be the sum of its n terms. Then,

$$T_r = (2r+1)r^2, \quad r=1, 2, 3, \dots, n$$

$$\therefore S_n = \sum_{r=1}^n T_r = \sum_{r=1}^n (2r+1)r^2 = \sum_{r=1}^n (2r^3 + r^2) = 2 \sum_{r=1}^n r^3 + \sum_{r=1}^n r^2$$

$$\Rightarrow S_n = 2 \left\{ \frac{n(n+1)}{2} \right\}^2 + \frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)(3n^2+5n+1)}{6}$$

10. (iv) We have,

$$T_n = n(n+1)(n+4) = n^3 + 5n^2 + 4n$$

Let S_n be the sum of n terms. Then,

$$\therefore S_n = \sum_{r=1}^n T_r = \sum_{r=1}^n (r^3 + 5r^2 + 4r)$$

$$\Rightarrow S_n = \sum_{r=1}^n r^3 + 5 \sum_{r=1}^n r^2 + 4 \sum_{r=1}^n r$$

$$\Rightarrow S_n = \left\{ \frac{n(n+1)}{2} \right\}^2 + \frac{5n(n+1)(2n+1)}{6} + 4 \frac{n(n+1)}{2} = \frac{n(n+1)(3n^2+23n+34)}{12}$$

(v) We have, $T_n = (2n-1)^2$

Let S_n be the sum of n terms of the given series. Then,

$$\therefore S_n = \sum_{r=1}^n T_r = \sum_{r=1}^n (2r-1)^2 = \sum_{r=1}^n (4r^2 - 4r + 1)$$

$$\Rightarrow S_n = 4 \sum_{r=1}^n r^2 - 4 \sum_{r=1}^n r + \sum_{r=1}^n 1$$

$$\Rightarrow S_n = 4 \frac{n(n+1)(2n+1)}{6} - 4 \frac{n(n+1)}{2} + n = \frac{n(2n-1)(2n+1)}{3}$$

11. Let T_r be the r^{th} term of the given series. Then,

$$T_r = 2r(2r+2), r=1, 2, 3, \dots$$

$$\Rightarrow T_r = 4r^2 + 4r$$

$$\therefore T_{20} = 4 \times 20^2 + 4 \times 20 = 1600 + 80 = 1680$$

Let S_{20} be the sum of 20 terms. Then,

$$S_{20} = \sum_{r=1}^{20} T_r = \sum_{r=1}^{20} (4r^2 + 4r) = 4 \left(\sum_{r=1}^{20} r^2 \right) + \left(\sum_{r=1}^{20} r \right)$$

$$\Rightarrow S_{20} = 4 \times \frac{20(20+1)(40+1)}{6} + 4 \times \frac{20(20+1)}{2} = 12320$$

21.2 METHOD OF DIFFERENCE

Sometimes the n^{th} term of a series can not be determined by the methods discussed so far. If a series is such that the difference between successive terms are either in A.P. or in G.P., then we determine its n^{th} term by the method of difference and then find the sum of the series by using the formulas for Σn , Σn^2 and Σn^3 . The method of difference is illustrated in the following examples.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 Find the sum to n terms of the series: $3 + 15 + 35 + 63 + \dots$

SOLUTION The difference between the successive terms are $15 - 3 = 12$, $35 - 15 = 20$, $63 - 35 = 28$, Clearly, these differences are in A.P.

Let T_n be the n^{th} term and S_n denote the sum to n terms of the given series. Then,

$$S_n = 3 + 15 + 35 + 63 + \dots + T_{n-1} + T_n \quad \dots(i)$$

$$\text{Also, } S_n = 3 + 15 + 35 + \dots + T_{n-1} + T_n \quad \dots(ii)$$

Subtracting (ii) from (i), we get

$$0 = 3 + \left\{ 12 + 20 + 28 + \dots + (T_n - T_{n-1}) \right\} - T_n$$

$$\Rightarrow T_n = 3 + \frac{(n-1)}{2} \left\{ 2 \times 12 + (n-1-1) \times 8 \right\} = 3 + (n-1)(12 + 4n - 8)$$

$$\Rightarrow T_n = 3 + (n-1)(4n+4) = 4n^2 - 1$$

$$\therefore S_n = \sum_{k=1}^n T_k = \sum_{k=1}^n (4k^2 - 1)$$

$$\Rightarrow S_n = 4 \sum_{k=1}^n k^2 - \sum_{k=1}^n 1 = 4 \left\{ \frac{n(n+1)(2n+1)}{6} \right\} - n = \frac{n}{3} (4n^2 + 6n - 1)$$

REMARK Instead of determining the n th term of a series by the method of difference as discussed in the above example, we can use the following steps to obtain the same.

STEP I Obtain the terms of the series and compute the differences between $T_2 - T_1, T_3 - T_2, T_4 - T_3, \dots$ etc. If these are in A.P., then take the n th term as $T_n = an^2 + bn + c$, where a, b, c are constants. Determine constants a, b, c by putting $n = 1, 2, 3$ and equating them with the values of corresponding terms of the given series.

STEP II If the differences $T_2 - T_1, T_3 - T_2, T_4 - T_3, \dots$ are in G.P. with common ratio r , then take $T_n = ar^{n-1} + bn + c$ and determine constants by putting $n = 1, 2, 3$ in T_n .

STEP III If the differences of the differences computed in step I are in A.P., then take $T_n = an^3 + bn^2 + cn + d$ and find the values of a, b, c, d by putting $n = 1, 2, 3, 4$.

STEP IV If the differences of the differences computed in step I are in G.P. with common ratio r , then take $T_n = ar^{n-1} + bn^2 + cn + d$ and find the values of a, b, c, d by putting $n = 1, 2, 3, 4$.

EXAMPLE 2 Find the sum to n terms of the series: $1 + 5 + 12 + 22 + 35 + \dots$

SOLUTION The sequence of differences between successive terms is $4, 7, 10, 13, \dots$, which is clearly an A.P. Let T_n be the n th term of the sequence and S_n be the sum of its n terms. Then,

$$S_n = 1 + 5 + 12 + 22 + 35 + \dots + T_{n-1} + T_n \quad \dots(i)$$

$$\text{Also, } S_n = 1 + 5 + 12 + 22 + \dots + T_{n-1} + T_n \quad \dots(ii)$$

Subtracting (ii) from (i), we get

$$0 = 1 + \{4 + 7 + 10 + 13 + \dots + (T_n - T_{n-1})\} - T_n$$

$$\Rightarrow T_n = 1 + \frac{(n-1)}{2} \{2 \times 4 + (n-1-1) \times 3\} = 1 + \left(\frac{n-1}{2}\right)(3n+2)$$

$$\Rightarrow T_n = \frac{1}{2}(3n^2 - n)$$

$$\therefore S_n = \sum_{k=1}^n T_k = \sum_{k=1}^n \frac{1}{2}(3k^2 - k) = \frac{3}{2} \sum_{k=1}^n k^2 - \frac{1}{2} \sum_{k=1}^n k$$

$$\Rightarrow S_n = \frac{3}{2} \left\{ \frac{n(n+1)(2n+1)}{6} \right\} - \frac{1}{2} \left\{ \frac{n(n+1)}{2} \right\} = \frac{n^2(n+1)}{2}$$

ALITER The given series is : $1 + 5 + 12 + 22 + 35 + \dots$

The sequence of differences between successive terms is : $4, 7, 10, 13, \dots$

Clearly, it is an A.P. So, let the n th term T_n of the given series be

$$T_n = an^2 + bn + c \quad \dots(i)$$

Putting $n = 1, 2, 3$, we get

$$T_1 = a + b + c \Rightarrow a + b + c = 1 \quad [\because T_1 = 1]$$

$$T_2 = 4a + 2b + c \Rightarrow 4a + 2b + c = 5 \quad [\because T_2 = 5]$$

$$T_3 = 9a + 3b + c \Rightarrow 9a + 3b + c = 12 \quad [\because T_3 = 12]$$

Solving these equations, we get

$$a = \frac{3}{2}, b = -\frac{1}{2} \text{ and } c = 0$$

Substituting the values of a, b, c in (i), we get

$$T_n = \frac{3}{2}n^2 - \frac{1}{2}n = \frac{1}{2}(3n^2 - n)$$

$$\begin{aligned} \therefore \text{Sum of the given series} &= \sum_{r=1}^n T_r = \sum_{r=1}^n \frac{1}{2} (3r^2 - r) = \frac{3}{2} \sum_{r=1}^n r^2 - \frac{1}{2} \sum_{r=1}^n r \\ &= \frac{3}{2} \left\{ \frac{n(n+1)(2n+1)}{6} \right\} - \frac{1}{2} \left\{ \frac{n(n+1)}{2} \right\} = \frac{n^2(n+1)}{2} \end{aligned}$$

EXAMPLE 3 Find the sum of first n terms of the following series:

(i) $3 + 7 + 13 + 21 + 31 + \dots$

[NCERT]

(ii) $5 + 11 + 19 + 29 + 41 + \dots$

[NCERT]

SOLUTION (i) The given series is: $3 + 7 + 13 + 21 + 31 + \dots$

The sequence of the differences between the successive terms of this series is

$$4, 6, 8, 10, \dots$$

Clearly, it is an A.P. with common difference 2. So, let the n th term of the given series be

$$T_n = an^2 + bn + c \quad \dots(i)$$

Putting $n=1, 2, 3$, we get

$$T_1 = a + b + c \Rightarrow a + b + c = 3 \quad [\because T_1 = 3]$$

$$T_2 = 4a + 2b + c \Rightarrow 4a + 2b + c = 7 \quad [\because T_2 = 7]$$

$$T_3 = 9a + 3b + c \Rightarrow 9a + 3b + c = 13 \quad [\because T_3 = 13]$$

Solving these equations, we get: $a = b = c = 1$

$$\therefore T_n = n^2 + n + 1$$

$$\begin{aligned} \text{Required sum} &= \sum_{r=1}^n T_r = \sum_{r=1}^n (r^2 + r + 1) = \sum_{r=1}^n r^2 + \sum_{r=1}^n r + \sum_{r=1}^n 1 \\ &= \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} + n = \frac{n}{3} (n^2 + 3n + 5) \end{aligned}$$

(ii) The given series is: $5 + 11 + 19 + 29 + 41 + \dots$

The sequence of the differences between the successive terms is: $6, 8, 10, 12, \dots$

Clearly, it is an A.P. So, n th term of the given series is given by

$$T_n = an^2 + bn + c \quad \dots(ii)$$

Putting $n=1, 2, 3$, we get

$$T_1 = a + b + c \Rightarrow a + b + c = 5 \quad [\because T_1 = 5]$$

$$T_2 = 4a + 2b + c \Rightarrow 4a + 2b + c = 11 \quad [\because T_2 = 11]$$

$$T_3 = 9a + 3b + c \Rightarrow 9a + 3b + c = 19 \quad [\because T_3 = 19]$$

Solving these three equations, we get: $a = 1, b = 3$ and $c = 1$.

$$\therefore T_n = n^2 + 3n + 1$$

$$\begin{aligned} \text{Required sum} &= \sum_{r=1}^n T_r = \sum_{r=1}^n (r^2 + 3r + 1) = \sum_{r=1}^n r^2 + 3 \sum_{r=1}^n r + \sum_{r=1}^n 1 \\ &= \frac{n(n+1)(2n+1)}{6} + \frac{3n(n+1)}{2} + n = \frac{n(n+2)(n+4)}{3} \end{aligned}$$

LEVEL-2

EXAMPLE 4 Sum the following series to n terms: $5 + 7 + 13 + 31 + 85 + \dots$

SOLUTION The sequence of differences between successive terms is 2, 6, 18, 54, ...

Clearly, it is a G.P. Let T_n be the n th term of the given series and S_n be the sum of its n terms. Then,

$$S_n = 5 + 7 + 13 + 31 + 85 + \dots + T_{n-1} + T_n \quad \dots(i)$$

$$\text{Also, } S_n = 5 + 7 + 13 + 31 + \dots + T_{n-1} + T_n \quad \dots(ii)$$

Subtracting (ii) from (i), we get

$$0 = 5 + \left\{ 2 + 6 + 18 + 54 + \dots + (T_n - T_{n-1}) \right\} - T_n$$

$$\Rightarrow 0 = 5 + \frac{2(3^{n-1} - 1)}{(3 - 1)} - T_n$$

$$\Rightarrow T_n = 5 + (3^{n-1} - 1) = 4 + 3^{n-1}$$

$$\therefore S_n = \sum_{k=1}^n T_k = \sum_{k=1}^n (4 + 3^{k-1}) = \sum_{k=1}^n 4 + \sum_{k=1}^n 3^{k-1}$$

$$\Rightarrow S_n = 4n + (1 + 3 + 3^2 + \dots + 3^{n-1})$$

$$\Rightarrow S_n = 4n + 1 \times \left(\frac{3^n - 1}{3 - 1} \right) = 4n + \left(\frac{3^n - 1}{2} \right) = \frac{1}{2} (3^n + 8n - 1)$$

ALITER The given series is: $5 + 7 + 13 + 31 + 85 + \dots$

The sequence of the differences between successive terms is 2, 6, 18, 54, ...

Clearly, it is a G.P. with common ratio 3. So, let the n th term of the given series be

$$T_n = a \cdot 3^{n-1} + bn + c \quad \dots(i)$$

Putting $n = 1, 2, 3$, we get

$$T_1 = a + b + c \Rightarrow a + b + c = 5 \quad [\because T_1 = 5]$$

$$T_2 = 3a + 2b + c \Rightarrow 3a + 2b + c = 7 \quad [\because T_2 = 7]$$

$$T_3 = 9a + 3b + c \Rightarrow 9a + 3b + c = 13 \quad [\because T_3 = 13]$$

Solving these equations, we get: $a = 1, b = 0$ and $c = 4$

Substituting the values of a, b, c in (i), we get

$$T_n = 3^{n-1} + 4$$

$$\begin{aligned} \therefore \text{Sum of the series} &= \sum_{r=1}^n T_r = \sum_{r=1}^n (3^{r-1} + 4) = \sum_{r=1}^n 3^{r-1} + \sum_{r=1}^n 4 \\ &= (1 + 3 + 3^2 + \dots + 3^{n-1}) + 4n \\ &= \frac{3^n - 1}{3 - 1} + 4n = \frac{3^n - 1}{2} + 4n = \frac{1}{2} (3^n - 1 + 8n) \end{aligned}$$

21.3 SUMMATION OF SOME SPECIAL SERIES

In this section, we shall discuss some problems for finding the sum of some series of the form

$$\frac{1}{a(a+d)} + \frac{1}{(a+d)(a+2d)} + \frac{1}{(a+2d)(a+3d)} + \dots + \frac{1}{(a+(n-2)d)(a+(n-1)d)}$$

In order to find the sum of a finite number of terms of such series, we write its each term as the difference of two terms as given below

$$\frac{1}{a(a+d)} = \frac{1}{d} \left(\frac{1}{a} - \frac{1}{a+d} \right),$$

$$\frac{1}{(a+d)(a+2d)} = \frac{1}{d} \left(\frac{1}{a+d} - \frac{1}{a+2d} \right),$$

$$\frac{1}{(a+2d)(a+3d)} = \frac{1}{d} \left(\frac{1}{a+2d} - \frac{1}{a+3d} \right) \text{ and so on.}$$

$$\begin{aligned} \therefore \frac{1}{a(a+d)} + \frac{1}{(a+d)(a+2d)} + \frac{1}{(a+2d)(a+3d)} + \dots + \frac{1}{\{a+(n-2)d\}\{a+(n-1)d\}} \\ = \frac{1}{d} \left\{ \left(\frac{1}{a} - \frac{1}{a+d} \right) + \left(\frac{1}{a+d} - \frac{1}{a+2d} \right) + \left(\frac{1}{a+2d} - \frac{1}{a+3d} \right) + \dots + \left(\frac{1}{a+(n-2)d} - \frac{1}{a+(n-1)d} \right) \right\} \\ = \frac{1}{d} \left\{ \frac{1}{a} - \frac{1}{a+(n-1)d} \right\} = \frac{n-1}{a\{a+(n-1)d\}} \end{aligned}$$

Following examples will illustrate the above procedure.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 Find the sum to n terms of the series:

$$\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n.(n+1)}$$

[NCERT]

SOLUTION Let T_r be the r th term of the given series. Then,

$$T_r = \frac{1}{r(r+1)}, r = 1, 2, \dots, n$$

$$\Rightarrow T_r = \frac{1}{r} - \frac{1}{r+1}, r = 1, 2, \dots, n$$

$$\begin{aligned} \therefore \text{Required sum} &= \sum_{r=1}^n T_r = \sum_{r=1}^n \left(\frac{1}{r} - \frac{1}{r+1} \right) \\ &= \left(1 - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \dots + \left(\frac{1}{n} - \frac{1}{n+1} \right) = 1 - \frac{1}{n+1} \end{aligned}$$

EXAMPLE 2 Find the sum to n terms of the series: $\frac{1}{1.3} + \frac{1}{3.5} + \frac{1}{5.7} + \dots$

SOLUTION Let T_r be the r th term of the given series. Then,

$$T_r = \frac{1}{(2r-1)(2r+1)}, r = 1, 2, 3, \dots, n$$

$$\Rightarrow T_r = \frac{1}{2} \left(\frac{1}{2r-1} - \frac{1}{2r+1} \right), r = 1, 2, 3, \dots, n$$

$$\therefore \text{Required sum} = \sum_{r=1}^n T_r = \frac{1}{2} \left\{ \left(\frac{1}{1} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{7} \right) + \dots + \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right) \right\}$$

$$= \frac{1}{2} \left\{ 1 - \frac{1}{2n+1} \right\} = \frac{n}{2n+1}$$

EXAMPLE 3 Find the sum : $\sum_{r=1}^n \frac{1}{(ar+b)(ar+a+b)}$

SOLUTION We have,

$$\begin{aligned} & \sum_{r=1}^n \frac{1}{(ar+b)(ar+a+b)} \\ &= \sum_{r=1}^n \frac{1}{a} \left(\frac{1}{ar+b} - \frac{1}{ar+a+b} \right) \\ &= \frac{1}{a} \sum_{r=1}^n \left(\frac{1}{ar+b} - \frac{1}{ar+a+b} \right) \\ &= \frac{1}{a} \left[\left(\frac{1}{a+b} - \frac{1}{2a+b} \right) + \left(\frac{1}{2a+b} - \frac{1}{3a+b} \right) + \dots + \left(\frac{1}{na+b} - \frac{1}{(n+1)a+b} \right) \right] \\ &= \frac{1}{a} \left\{ \frac{1}{a+b} - \frac{1}{(n+1)a+b} \right\} = \frac{n}{(a+b) \{ (n+1)a+b \}} \end{aligned}$$

LEVEL-2

EXAMPLE 4 Find the sum to n terms of the series: $\frac{3}{1^2 \cdot 2^2} + \frac{5}{2^2 \cdot 3^2} + \frac{7}{3^2 \cdot 4^2} + \dots$

SOLUTION Let T_r be the r th term of the given series. Then,

$$T_r = \frac{(2r+1)}{r^2(r+1)^2}, r = 1, 2, 3, \dots$$

$$\Rightarrow T_r = \frac{(r+1)^2 - r^2}{r^2(r+1)^2}, r = 1, 2, 3, \dots$$

$$\Rightarrow T_r = \left\{ \frac{1}{r^2} - \frac{1}{(r+1)^2} \right\}, r = 1, 2, 3, \dots$$

Let S_n be the sum to n terms of the given series. Then,

$$\therefore \text{Required sum} = \sum_{r=1}^n T_r = \sum_{r=1}^n \left\{ \frac{1}{r^2} - \frac{1}{(r+1)^2} \right\} = 1 - \frac{1}{(n+1)^2} = \frac{2n+n^2}{(n+1)^2}$$

EXAMPLE 5 Find the sum to n terms of the series:

$$\frac{1}{1+1^2+1^4} + \frac{2}{1+2^2+2^4} + \frac{3}{1+3^2+3^4} + \dots$$

SOLUTION Let T_r be the r th term of the given series. Then,

$$T_r = \frac{r}{1+r^2+r^4}, r = 1, 2, 3, \dots, n$$

$$\Rightarrow T_r = \frac{r}{(r^2+r+1)(r^2-r+1)} = \frac{1}{2} \left\{ \frac{2r}{(r^2+r+1)(r^2-r+1)} \right\} = \frac{1}{2} \left\{ \frac{(r^2+r+1)-(r^2-r+1)}{(r^2+r+1)(r^2-r+1)} \right\}$$

$$\Rightarrow T_r = \frac{1}{2} \left\{ \frac{1}{r^2 - r + 1} - \frac{1}{r^2 + r + 1} \right\}, r = 1, 2, \dots, n$$

Let S_n be the sum to n terms of the given series. Then,

$$\begin{aligned} \therefore S_n &= \sum_{r=1}^n T_r = \frac{1}{2} \left\{ \sum_{r=1}^n \left(\frac{1}{r^2 - r + 1} - \frac{1}{r^2 + r + 1} \right) \right\} \\ &= \frac{1}{2} \left\{ \left(1 - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{7} \right) + \left(\frac{1}{7} - \frac{1}{13} \right) + \dots + \left(\frac{1}{n^2 - n + 1} - \frac{1}{n^2 + n + 1} \right) \right\} \\ &= \frac{1}{2} \left\{ 1 - \frac{1}{n^2 + n + 1} \right\} = \frac{n^2 + n}{2(n^2 + n + 1)} \end{aligned}$$

EXERCISE 21.2**LEVEL-1**

Sum the following series to n terms:

1. $3 + 5 + 9 + 15 + 23 + \dots$
2. $2 + 5 + 10 + 17 + 26 + \dots$
3. $1 + 3 + 7 + 13 + 21 + \dots$
4. $3 + 7 + 14 + 24 + 37 + \dots$
5. $1 + 3 + 6 + 10 + 15 + \dots$
6. $1 + 4 + 13 + 40 + 121 + \dots$
7. $4 + 6 + 9 + 13 + 18 + \dots$
8. $2 + 4 + 7 + 11 + 16 + \dots$
9. $\frac{1}{1.4} + \frac{1}{4.7} + \frac{1}{7.10} + \dots$
10. $\frac{1}{1.6} + \frac{1}{6.11} + \frac{1}{11.16} + \frac{1}{16.21} + \dots + \frac{1}{(5n-4)(5n+1)}$

ANSWERS

1. $\frac{n}{3}(n^2 + 8)$
2. $\frac{n}{6}(2n^2 + 3n + 7)$
3. $\frac{n(n^2 + 2)}{3}$
4. $\frac{n}{2}(n^2 + n + 4)$
5. $\frac{n}{6}(n+1)(n+2)$
6. $\frac{1}{4}(3^{n+1} - 2n - 3)$
7. $\frac{n}{6}(n^2 + 3n + 20)$
8. $\frac{n}{6}(n^2 + 3n + 8)$
9. $\frac{n}{3n+1}$
10. $\frac{n}{5n+1}$

VERY SHORT ANSWER QUESTIONS (VSAQs)

Answer each of the following questions in one word or one sentence or as per exact requirement of the question:

1. Write the sum of the series: $2 + 4 + 6 + 8 + \dots + 2n$.
2. Write the sum of the series: $1^2 - 2^2 + 3^2 - 4^2 + 5^2 - 6^2 + \dots + (2n-1)^2 - (2n)^2$.

- Write the sum to n terms of a series whose r^{th} term is: $r + 2^r$.
- If $\sum_{r=1}^n r = 55$, find $\sum_{r=1}^n r^3$.
- If the sum of first n even natural numbers is equal to k times the sum of first n odd natural numbers, then write the value of k .
- Write the sum of 20 terms of the series: $1 + \frac{1}{2}(1+2) + \frac{1}{3}(1+2+3) + \dots$
- Write the 50th term of the series $2 + 3 + 6 + 11 + 18 + \dots$
- Let S_n denote the sum of the cubes of first n natural numbers and s_n denote the sum of first n natural numbers. Then, write the value of $\sum_{r=1}^n \frac{S_r}{s_r}$.

ANSWERS

- $n(n+1)$
- $-n(2n+1)$
- $\frac{n(n+1)}{2} + 2^{n+1} - 2$
- 3025
- $\frac{n+1}{n}$
- 115
- $49^2 + 2$
- $\frac{n(n+1)(n+2)}{6}$

MULTIPLE CHOICE QUESTIONS (MCQs)

Mark the correct alternative in each of the following:

- The sum to n terms of the series $\frac{1}{\sqrt{1} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{5}} + \frac{1}{\sqrt{5} + \sqrt{7}} + \dots$ is
 (a) $\sqrt{2n+1}$ (b) $\frac{1}{2}\sqrt{2n+1}$ (c) $\sqrt{2n+1} - 1$ (d) $\frac{1}{2}\left\{\sqrt{2n+1} - 1\right\}$
- The sum of the series: $\frac{1}{\log_2 4} + \frac{1}{\log_4 4} + \frac{1}{\log_8 4} + \dots + \frac{1}{\log_{2^n} 4}$ is
 (a) $\frac{n(n+1)}{2}$ (b) $\frac{n(n+1)(2n+1)}{12}$ (c) $\frac{n(n+1)}{4}$ (d) none of these
- The value of $\sum_{r=1}^n \left\{ (2r-1)a + \frac{1}{b^r} \right\}$ is equal to
 (a) $an^2 + \frac{b^{n-1}-1}{b^{n-1}(b-1)}$ (b) $an^2 + \frac{b^n-1}{b^n(b-1)}$
 (c) $an^3 + \frac{b^{n-1}-1}{b^n(b-1)}$ (d) none of these
- If $\Sigma n = 210$, then $\Sigma n^2 =$
 (a) 2870 (b) 2160 (c) 2970 (d) none of these
- If $S_n = \sum_{r=1}^n \frac{1+2+2^2+\dots+\text{Sum to } r \text{ terms}}{2^r}$, then S_n is equal to
 (a) $2^n - n - 1$ (b) $1 - \frac{1}{2^n}$ (c) $n - 1 + \frac{1}{2^n}$ (d) $2^n - 1$

6. If $1 + \frac{1+2}{2} + \frac{1+2+3}{3} + \dots$ to n terms is S . Then, S is equal to
 (a) $\frac{n(n+3)}{4}$ (b) $\frac{n(n+2)}{4}$ (c) $\frac{n(n+1)(n+2)}{6}$ (d) n^2
7. Sum of n terms of the series $\sqrt{2} + \sqrt{8} + \sqrt{18} + \sqrt{32} + \dots$ is
 (a) $\frac{n(n+1)}{2}$ (b) $2n(n+1)$ (c) $\frac{n(n+1)}{\sqrt{2}}$ (d) 1
8. The sum of 10 terms of the series $\sqrt{2} + \sqrt{6} + \sqrt{18} + \dots$ is
 (a) $121(\sqrt{6} + \sqrt{2})$ (b) $243(\sqrt{3} + 1)$ (c) $\frac{121}{\sqrt{3}-1}$ (d) $242(\sqrt{3}-1)$
9. The sum of the series $1^2 + 3^2 + 5^2 + \dots$ to n terms is
 (a) $\frac{n(n+1)(2n+1)}{2}$ (b) $\frac{n(2n-1)(2n+1)}{3}$ (c) $\frac{(n-1)^2(2n+1)}{6}$ (d) $\frac{(2n+1)^3}{3}$
10. The sum of the series $\frac{2}{3} + \frac{8}{9} + \frac{26}{27} + \frac{80}{81} + \dots$ to n terms is
 (a) $n - \frac{1}{2}(3^{-n} - 1)$ (b) $n - \frac{1}{2}(1 - 3^{-n})$ (c) $n + \frac{1}{2}(3^n - 1)$ (d) $n - \frac{1}{2}(3^n - 1)$

ANSWERS

1. (d) 2. (c) 3. (b) 4. (a) 5. (c) 6. (a) 7. (c) 8. (a)
 9. (b) 10. (b)

SUMMARY

1. For any $n \in N$, we have

$$\begin{aligned} \text{(i)} \quad \sum_{k=1}^n k &= 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2} \\ \text{(ii)} \quad \sum_{k=1}^n k^2 &= 1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6} \\ \text{(iii)} \quad \sum_{k=1}^n k^3 &= 1^3 + 2^3 + 3^3 + \dots + n^3 = \left\{ \frac{n(n+1)}{2} \right\}^2 \\ \text{(iv)} \quad \sum_{k=1}^n k^4 &= 1^4 + 2^4 + 3^4 + \dots + n^4 = \frac{n(n+1)(2n+1)(3n^2+3n-1)}{30} \end{aligned}$$

2. In a series $a_1 + a_2 + a_3 + a_4 + \dots + a_n + \dots$,

- (i) if the differences $a_2 - a_1, a_3 - a_2, a_4 - a_3, \dots$ are in A.P., then the n th term is given by
 $a_n = an^2 + bn + c$, where a, b, c are constants.
- (ii) if the differences $a_2 - a_1, a_3 - a_2, a_4 - a_3, \dots$ are in G.P. with common ratio r , then
 $a_n = ar^{n-1} + bn + c$, where a, b, c are constants.

To determine constants a, b, c we put $n=1, 2, 3$ and equate them with the values of corresponding terms of the given series.

CHAPTER 22

BRIEF REVIEW OF CARTESIAN SYSTEM OF RECTANGULAR CO-ORDINATES

22.1 CARTESIAN CO-ORDINATE SYSTEM

RECTANGULAR CO-ORDINATE AXES Let $X'OX$ and $Y'OY$ be two mutually perpendicular lines through any point O in the plane of the paper. We call the point O , the origin. Now choose a convenient unit of length and starting from the origin as zero, mark off a number scale on the horizontal line $X'OX$, positive to the right of the origin O and negative to the left of origin O . Also, mark off the same scale on the vertical line $Y'OY$, positive upwards and negative downwards of the origin O .

The line $X'OX$ is called the x -axis or axis of x , the line $Y'OY$ is known as the y -axis or axis of y , and the two lines taken together are called the co-ordinate axes or the axes of coordinates.

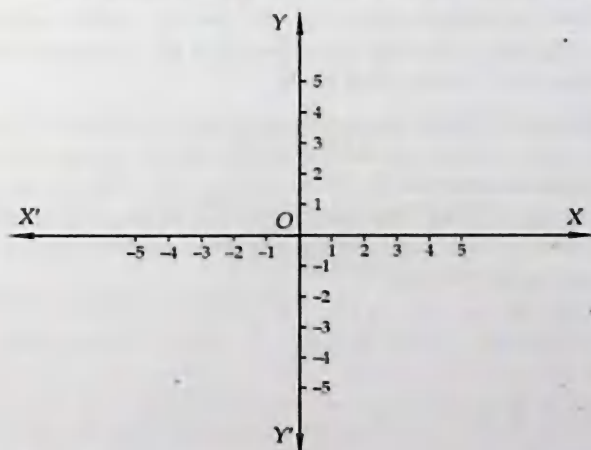


Fig. 22.1

CARTESIAN CO-ORDINATES OF A POINT Let $X'OX$ and $Y'OY$ be the co-ordinate axes, and let P be any point in the plane. Draw perpendiculars PM and PN from P on x and y -axis respectively. The length of the directed line segment OM in the units of scale chosen is called the x -coordinate or *abscissa* of point P . Similarly, the length of the directed line segment ON on the same scale is called the y -coordinate or *ordinate* of point P . Let $OM = x$ and $ON = y$. Then the position of the point P in the plane with respect to the coordinate axes is represented by the ordered (x, y) . The ordered pair (x, y) is called the coordinates of point P .

Thus, for a given point, the abscissa and ordinate are the distances of the given point from y -axis and x -axis respectively.

The above system of coordinating an ordered pair (x, y) with every point in a plane is called the *Rectangular Cartesian coordinate system*.

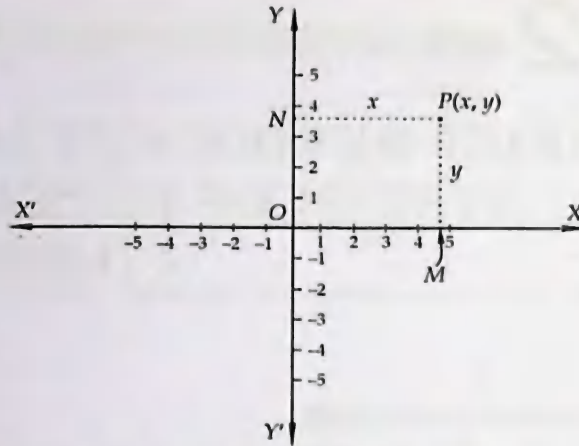


Fig. 22.2

It follows from the above discussion that corresponding to every point P in the Euclidean plane there is a unique ordered pair (x, y) of real numbers called its Cartesian coordinates. Conversely, when we are given an ordered pair (x, y) and a Cartesian coordinate system, we can determine a point in the Euclidean plane having its coordinates (x, y) . For this we mark off a directed line segment $OM = x$ on the x -axis and another directed line segment $ON = y$ on y -axis. Now, draw perpendiculars at M and N to X and Y axes respectively. The point of intersection of these two perpendiculars determines point P in the Euclidean space having coordinates (x, y) .

Thus, there is one-to-one correspondence between the set of all ordered pairs (x, y) of real numbers and the points in the Euclidean plane. The set of all ordered pairs (x, y) of real numbers is called the Cartesian plane and is denoted by R^2 .

QUADRANTS Let $X'OX$ and $Y'OY$ be the coordinate axes. We observe that the two axes divide the Euclidean plane into four regions, called the quadrants. The regions XOY , $X'OY$, $X'OY'$ and $Y'OX$ are known as the first, the second, the third and the fourth quadrants respectively. The ray OX is taken as positive x -axis, OX' as negative x -axis, OY as positive y -axis and OY' as negative y -axis. In view of the above sign convention the four quadrants are characterised by the following signs of abscissa and ordinate.

- I Quadrant: $x > 0, y > 0$
- II Quadrant: $x < 0, y > 0$
- III Quadrant: $x < 0, y < 0$
- IV Quadrant: $x > 0, y < 0$

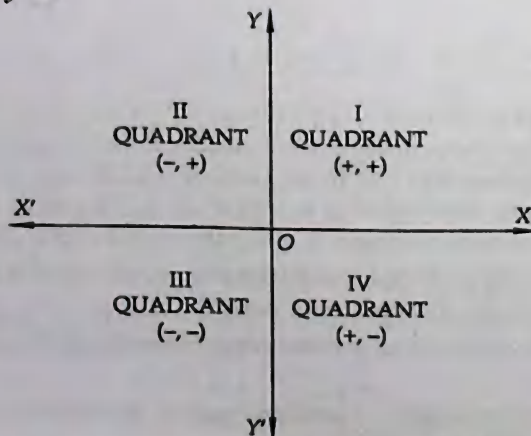


Fig. 22.3

The coordinates of the origin are taken as $(0, 0)$. The coordinates of any point on x -axis are of the form $(x, 0)$ and the coordinates of any point on y -axis are of the form $(0, y)$. Thus, if the abscissa of a point is zero, it would lie somewhere on the y -axis and if its ordinate is zero it would lie on x -axis.

It follows from the above discussion that by simply looking at the coordinates of a point we can tell in which quadrant it would lie.

22.2 DISTANCE BETWEEN TWO POINTS

The distance between any two points in the plane is the length of the line segment joining them.

The distance between two points $P(x_1, y_1)$ and $Q(x_2, y_2)$ is given by

$$PQ = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$\text{i.e. } PQ = \sqrt{(\text{Difference of abscisse})^2 + (\text{Difference of ordinates})^2}$$

SOME USEFUL POINTS

- (I) In order to prove that a given figure is a
- square, prove that the four sides are equal and the diagonals are also equal.
 - rhombus, prove that the four sides are equal.
 - rectangle, prove that opposite sides are equal and the diagonals are also equal.
 - a parallelogram, prove that the opposite sides are equal.
 - parallelogram but not a rectangle, prove that its opposite sides are equal but the diagonals are not equal.
 - a rhombus but not a square, prove that its all sides are equal but the diagonals are not equal.
- (II) For three points to be collinear, prove that the sum of the distances between two pairs of points is equal to the third pair of points.

ILLUSTRATIVE EXAMPLES

LEVEL-1

EXAMPLE 1 Find the distance between the points

- (i) $A(at_1^2, 2at_1)$ and $B(at_2^2, 2at_2)$ (ii) $L(a \cos \alpha, a \sin \alpha)$ and $M(a \cos \beta, a \sin \beta)$

SOLUTION (i) Clearly,

$$AB = \sqrt{(at_2^2 - at_1^2)^2 + (2at_2 - 2at_1)^2} = \sqrt{a^2(t_2 - t_1)^2(t_2 + t_1)^2 + 4a^2(t_2 - t_1)^2}$$

$$\Rightarrow AB = a(t_2 - t_1) \sqrt{(t_2 + t_1)^2 + 4}$$

$$\begin{aligned} \text{(ii) } LM &= \sqrt{(a \cos \beta - a \cos \alpha)^2 + (a \sin \beta - a \sin \alpha)^2} \\ &= \sqrt{a^2(\cos \beta - \cos \alpha)^2 + a^2(\sin \beta - \sin \alpha)^2} \\ &= a \sqrt{(\cos \beta - \cos \alpha)^2 + (\sin \beta - \sin \alpha)^2} \\ &= a \sqrt{\cos^2 \beta + \cos^2 \alpha + \sin^2 \beta + \sin^2 \alpha - 2 \cos \alpha \cos \beta - 2 \sin \alpha \sin \beta} \\ &= a \sqrt{(\cos^2 \beta + \sin^2 \beta) + (\cos^2 \alpha + \sin^2 \alpha) - 2(\cos \alpha \cos \beta + \sin \alpha \sin \beta)} \\ &= a \sqrt{1 + 1 - 2 \cos(\alpha - \beta)} = a \sqrt{2[1 - \cos(\alpha - \beta)]} \\ &= a \sqrt{2 \times 2 \sin^2\left(\frac{\alpha - \beta}{2}\right)} = 2a \sin\left(\frac{\alpha - \beta}{2}\right) \end{aligned}$$

EXAMPLE 2 Show that four points $(0, -1)$, $(6, 7)$, $(-2, 3)$ and $(8, 3)$ are the vertices of a rectangle.

SOLUTION Let $A(0, -1)$, $B(6, 7)$, $C(-2, 3)$ and $D(8, 3)$ be the given points. Then,

$$AD = \sqrt{(8-0)^2 + (3+1)^2} = \sqrt{64+16} = 4\sqrt{5}$$

$$BC = \sqrt{(6+2)^2 + (7-3)^2} = \sqrt{64+16} = 4\sqrt{5}$$

$$AC = \sqrt{(-2-0)^2 + (3+1)^2} = \sqrt{4+16} = 2\sqrt{5}$$

and, $BD = \sqrt{(8-6)^2 + (3-7)^2} = \sqrt{4+16} = 2\sqrt{5}$

$\therefore AD = BC$ and $AC = BD$.

So, $ADBC$ is a parallelogram.

Now, $AB = \sqrt{(6-0)^2 + (7+1)^2} = \sqrt{36+64} = 10$ and, $CD = \sqrt{(8+2)^2 + (3-3)^2} = 10$.

Clearly, $AB^2 = AD^2 + DB^2$ and $CD^2 = CB^2 + BD^2$. Hence, $ADBC$ is a rectangle.

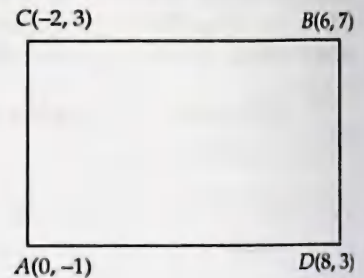


Fig. 22.4

LEVEL-2

EXAMPLE 3 If the two vertices of an equilateral triangle be $(0, 0)$, $(3, \sqrt{3})$, find the third vertex.

SOLUTION $O(0, 0)$ and $A(3, \sqrt{3})$ be the given points and let $B(x, y)$ be the third vertex of equilateral $\triangle OAB$. Then,

$$OA = OB = AB \Rightarrow OA^2 = OB^2 = AB^2 \quad \dots(i)$$

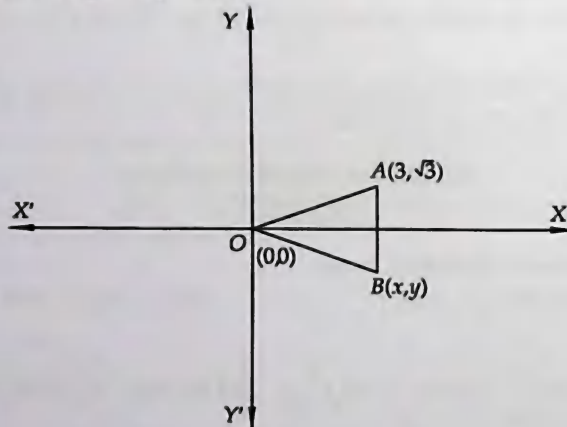


Fig. 22.5

Clearly, $OA^2 = (3-0)^2 + (\sqrt{3}-0)^2 = 12$, $OB^2 = x^2 + y^2$

and, $AB^2 = (x-3)^2 + (y-\sqrt{3})^2 = x^2 + y^2 - 6x - 2\sqrt{3}y + 12$

Now, $OA^2 = OB^2 = AB^2$

[From (i)]

$$\Rightarrow OA^2 = OB^2 \text{ and } OB^2 = AB^2$$

$$\Rightarrow x^2 + y^2 = 12 \text{ and } x^2 + y^2 = x^2 + y^2 - 6x - 2\sqrt{3}y + 12$$

$$\Rightarrow x^2 + y^2 = 12 \text{ and } 6x + 2\sqrt{3}y = 12$$

$$\Rightarrow x^2 + y^2 = 12 \text{ and } 3x + \sqrt{3}y = 6$$

$$\Rightarrow x^2 + \left(\frac{6-3x}{\sqrt{3}}\right)^2 = 12$$

$$\left[\because 3x + \sqrt{3}y = 6 \therefore y = \frac{6-3x}{\sqrt{3}} \right]$$